Assessments of the year

Organisation. Each deadline has two practice papers for training. They announce the expected methods; they may be given at the start of the unit, even if not all the topics have yet been taught. The table indicates the planned time of distribution. Versions A and B are reserved for the assessment.

Deadline	Administration guide	Practice papers
Test 1	S9 · week of 21 September	From S2
Test 2	S18 · week of 12 October	From S14
Common test	S28 · week of 16 November; slot to be harmonised	From S14
Test 3	S34 · week of 30 November	From S21
Mock exam 1	S45 · 14 to 16 January, 4 h slot to be fixed	From S36
Test 4	S57 · week of 8 February	From S44
Test 5	S68 · week of 22 March	From S52
Mock exam 2	S71 · 1st to 3 April, 4 h slot to be fixed	From S59
Final mock exam	S90 · week of 31 May, 4 h slot to be fixed	From S75

Test 1 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 1$ and, for every natural number n, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate u_1 and u_2. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number n, $1 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). *(1 pt)*
5. A student claims: "There exists an integer n such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate v_1 and v_2. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 9$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 5\vec{\jmath} + 9\vec{k}, \quad \vec{z} = 4\vec{\imath} + 5\vec{\jmath} + 10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $5\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(1 pt)*

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices E, F, G and H are the respective images of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI} = \frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines (IJ) and (BD) are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let O be the centre of the parallelepiped. Prove that O lies on the line (AG). *(1 pt)*
5. Prove that the points I, J and K are not collinear. *(1 pt)*

Test 1 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate u_1 and u_2. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number n, $2 \leq u_n < 9$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). *(1 pt)*
5. A student claims: "There exists an integer n such that $u_n = 9$." Is this claim true? Justify. *(0.5 pt)*

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate v_1 and v_2. Give the exact values. *(0.5 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. *(1.5 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(1.5 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 10$. *(1 pt)*
5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to conclude that $P(0)$ is true? Justify. *(0.5 pt)*

Test 1 · Practice 2 · continued

Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 7\vec{\jmath} + 10\vec{k}, \quad \vec{z} = 3\vec{\imath} + 7\vec{\jmath} + 11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $7\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(1 pt)*

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices E, F, G and H are the respective images of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI} = \frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines (IJ) and (BD) are parallel. *(1 pt)*
3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. *(1 pt)*
4. Let O be the centre of the parallelepiped. Prove that O lies on the line (AG). *(1 pt)*
5. Prove that the points I, J and K are not collinear. *(1 pt)*

Test 2 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ and $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of (u_n), detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. *(1 pt)*
3. Deduce the limit of (v_n). *(1 pt)*
4. We define $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of (w_n). *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. *(1 pt)*
2. The sequence $u_n = 2 + (-1)^n$ converges to 2. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. *(1 pt)*

Test 2 · Practice 1 · continued

Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 2\vec{\imath} + 6\vec{\jmath} + 8\vec{k}, \quad \vec{z} = 2\vec{\imath} + 6\vec{\jmath} + 9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $6\vec{u} - 2\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(1 pt)*

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices E, F, G and H are the respective images of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI} = \frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines (IJ) and (BD) are parallel. *(1 pt)*
3. Study the relative position of the lines (AB) and (CG): are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let O be the centre of the parallelepiped. Does the point O lie in the plane (ABG)? Justify. *(1 pt)*
5. Prove that the points I, J and K are not collinear. *(1 pt)*

Test 2 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ and $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of (u_n), detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. *(1 pt)*
3. Deduce the limit of (v_n). *(1 pt)*
4. We define $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of (w_n). *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, say whether it is true or false and justify your answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. *(1 pt)*
2. The sequence $u_n = 5 + (-1)^n$ converges to 5. *(1 pt)*
3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. *(1 pt)*
4. The sequence $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converges to 1. *(1 pt)*
5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. *(1 pt)*

Test 2 · Practice 2 · continued

Exercise 3 · Vector calculations in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, we define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 7\vec{\jmath} + 11\vec{k}, \quad \vec{z} = 4\vec{\imath} + 7\vec{\jmath} + 12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(1 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(1.5 pt)*
4. Express the vector $7\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(1 pt)*

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped $ABCDEFGH$ that is not flat. The vertices E, F, G and H are the respective images of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI} = \frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ} = \frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK} = \frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
2. Prove that the lines (IJ) and (BD) are parallel. *(1 pt)*
3. Study the relative position of the lines (AB) and (CG): are they parallel, intersecting or non-coplanar? Justify. *(1 pt)*
4. Let O be the centre of the parallelepiped. Does the point O lie in the plane (ABG)? Justify. *(1 pt)*
5. Prove that the points I, J and K are not collinear. *(1 pt)*

Common test · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate u_1 and u_2. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number n, $2 \leq u_n < 8$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). *(1 pt)*
5. A student claims: "There exists an integer n such that $u_n = 8$." Is this claim true? Justify. *(0.5 pt)*

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ and $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n), detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. *(1 pt)*
3. Deduce the limit of (v_n). *(1 pt)*
4. We define $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · Practice 1 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{2x+6}{x-2}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of f. Find two real numbers A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of f at 2^- and at 2^+. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ with justification. *(1 pt)*

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet containing 15 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

Common test · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 3$ and, for every natural number n, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate u_1 and u_2. *(0.5 pt)*
2. Prove by induction that, for all $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. *(2 pt)*
3. Deduce that, for every natural number n, $3 \leq u_n < 10$. *(1 pt)*
4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). *(1 pt)*
5. A student claims: "There exists an integer n such that $u_n = 10$." Is this claim true? Justify. *(0.5 pt)*

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ and $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of (u_n), detailing the transformation of its expression. *(1 pt)*
2. Show that, for every integer $n \geq 1$, $v_n = \dfrac{7}{\sqrt{1+\frac{7}{n}}+1}$. *(1 pt)*
3. Deduce the limit of (v_n). *(1 pt)*
4. We define $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of (w_n). *(1 pt)*
5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this claim true? Justify. *(1 pt)*

Common test · Practice 2 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+7}{x-3}$. Denote by $\mathcal{C}_f$ its graph in a coordinate frame of the plane.

1. Determine the domain of f. Find two real numbers A and B such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of f at 3^- and at 3^+. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ with justification. *(1 pt)*

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet containing 16 distinct symbols. The rules about repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? *(1 pt)*
2. How many codes can be formed if each symbol must be used at most once? *(1 pt)*
3. A symbol of the alphabet is fixed in advance. Repetitions being allowed, how many codes contain this symbol at least once? *(1 pt)*
4. The first symbol of the code is imposed and repetitions are forbidden. How many codes can be formed? *(1 pt)*
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? *(1 pt)*

Test 3 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+4}{x-1}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of f. Find two real numbers A and B such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of f at 1^- and at 1^+. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-1x}$ with justification. *(1 pt)*

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ par $f(x) = x^3 + 4x - 5$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of f, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · Practice 1 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0, +\infty)$ par $f(x) = (x+3)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of f. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of f. *(1 pt)*
4. Prove that the equation $f(x) = \frac{3}{2}$ has a unique solution α in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

Exercise 4 · Forming a group — 5 points

A group consists of 6 teachers and 10 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

Test 3 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{4x+6}{x-2}$. Let $\mathcal{C}_f$ denote its graph in a coordinate plane.

1. Determine the domain of f. Find two real numbers A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(1 pt)*
3. Determine the limits of f at 2^- and at 2^+. *(1 pt)*
4. Give the equations of the asymptotes of $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. *(1 pt)*
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^4 e^{-2x}$ with justification. *(1 pt)*

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ par $f(x) = x^3 + 3x - 7$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. *(1 pt)*
2. Determine the limits of f at $-\infty$ and at $+\infty$. *(0.5 pt)*
3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. *(1.5 pt)*
4. Study the convexity of f, then determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0, then study the position of the curve relative to this tangent. *(1 pt)*

Test 3 · Practice 2 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0, +\infty)$ par $f(x) = (x+2)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of f. *(1 pt)*
3. Calculate $f''(x)$, then study the convexity of f. *(1 pt)*
4. Prove that the equation $f(x) = 1$ has a unique solution α in $[0, +\infty)$. *(1.5 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(1 pt)*

Exercise 4 · Forming a group — 5 points

A group consists of 9 teachers and 11 students, all distinct. A team of four people is chosen; the order of selection does not matter.

1. How many teams of four people can be formed? *(1 pt)*
2. How many of these teams contain exactly two teachers? *(1 pt)*
3. How many of these teams contain at least one teacher and at least one student? *(1 pt)*
4. A team containing exactly two teachers is chosen, then among its members a teacher spokesperson and a student reporter are designated. Count the choices of the team together with these two assignments. *(1 pt)*
5. Explain why the number of ordered lists of four distinct people, divided by 4!, gives the number of teams obtained in question 1. *(1 pt)*

Mock 1 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 7$. *(0.5 pt)*
5. Justify that (v_n) converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $7 - v_n$ in terms of n. Then determine, by comparing integer powers, an index from which this bound guarantees $7 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer n such that $7 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0, +\infty)$ by $f(x) = (x + 5)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of f. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of f. *(0.5 pt)*
4. Prove that the equation $f(x) = \frac{5}{2}$ has a unique solution α in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of α of width at most 10^{-3}. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the horizontal axis." Refute this claim by studying the sign of $f(x)$. *(1 pt)*

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 4\vec{\imath} + 6\vec{\jmath} + 10\vec{k}, \quad \vec{z} = 4\vec{\imath} + 6\vec{\jmath} + 11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $6\vec{u} - 4\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. *(0.5 pt)*
7. Let I be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points A, I and D are collinear. *(0.5 pt)*
8. Define E by $\overrightarrow{AE} = \vec{z}$. Is the line (DE) contained in the plane (ABC)? Justify. *(1 pt)*

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 19 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 19\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 19 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

Mock 1 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. Give the exact values. *(0.25 pt)*
2. Prove by induction that all terms are defined and that, for all $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. *(0.75 pt)*
3. Prove by induction that, for all $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. *(0.75 pt)*
4. Prove that, for all $n \in \mathbb{N}$, $v_n < 11$. *(0.5 pt)*
5. Justify that (v_n) converges, then determine its limit. *(0.75 pt)*
6. Prove that, for all $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. *(0.5 pt)*
7. Deduce by induction an upper bound for $11 - v_n$ in terms of n. Then determine, by comparing integer powers, an index from which this bound guarantees $11 - v_n \leq 10^{-3}$. *(1 pt)*
8. Write a Python function with no argument that returns the first integer n such that $11 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. *(0.5 pt)*

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0, +\infty)$ by $f(x) = (x + 4)e^{-x}$. Let $\mathcal{C}_f$ denote its graph.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw the variation table of f. *(0.5 pt)*
3. Calculate $f''(x)$, then study the convexity of f. *(0.5 pt)*
4. Prove that the equation $f(x) = 2$ has a unique solution α in $[0, +\infty)$. *(1 pt)*
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point of abscissa 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. *(0.5 pt)*
6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0, 2]$ and returns a bracketing of α of width at most 10^{-3}. *(0.5 pt)*
7. Determine the minimum number of bisection steps guaranteeing this width. *(0.5 pt)*
8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the horizontal axis." Refute this claim by studying the sign of $f(x)$. *(1 pt)*

Mock 1 · Practice 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$ of space, define the vectors

$$\vec{u} = \vec{\imath} + \vec{k}, \qquad \vec{v} = \vec{\jmath} + \vec{k},$$
$$\vec{w} = 3\vec{\imath} + 8\vec{\jmath} + 11\vec{k}, \quad \vec{z} = 3\vec{\imath} + 8\vec{\jmath} + 12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. *(0.5 pt)*
2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. *(0.5 pt)*
3. Are the vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. *(0.5 pt)*
4. Express the vector $8\vec{u} - 3\vec{v}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$. *(1 pt)*
5. Choose a point A, then define B, C and D by $\overrightarrow{AB} = \vec{u}$, $\overrightarrow{AC} = \vec{v}$ and $\overrightarrow{AD} = \vec{w}$. Prove that A, B, C and D are coplanar. *(0.5 pt)*
6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. *(0.5 pt)*
7. Let I be the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$, then determine whether the points A, I and D are collinear. *(0.5 pt)*
8. Define E by $\overrightarrow{AE} = \vec{z}$. Is the line (DE) contained in the plane (ABC)? Justify. *(1 pt)*

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 16 times. The trials are independent and the probability of success is $p = \frac{1}{10}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Express using complementary events, then calculate, the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. The same trial is now repeated $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 16\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 16 trials is accepted if its number of successes is between 1 and 3 inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

Test 4 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{3+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of f at 0^+ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of f. *(1 pt)*
3. State the value of the maximum of f, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

Exercise 2 · Antiderivatives with a given condition — 5 points

The first four questions are independent. For each, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 2x^2 - 6x + 2$. *(1 pt)*
2. Determine the antiderivative F of $x \mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0) = 6$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+6}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+2)^2$. *(1 pt)*
5. Let F and G be two antiderivatives of the same function on an interval I. Prove that there exists a real constant C such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · Practice 1 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0, +\infty)$. In questions 2 to 5, denote by y the solution satisfying $y(0) = 13$.

1. Determine the constant solution of (E). *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as t tends to $+\infty$. *(1 pt)*
4. Study the variations of y, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 9$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by Y_k the approximation of $y(kh)$. Give the value of Y_0 and express Y_{k+1} in terms of Y_k and h. *(1 pt)*

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, consider $A(11, 0, 0)$, $B(0, 11, 0)$ and $C(0, 0, 11)$. We are given $\vec{n} = (1, 1, 1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane (ABC). *(1 pt)*
2. Let H be the orthogonal projection of O onto (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that OH is the distance from O to the plane (ABC): for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle ABC. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

Test 4 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{2+\ln x}{x}$. We denote by $\mathcal{C}_f$ its graph.

1. Determine the limits of f at 0^+ and at $+\infty$. *(1 pt)*
2. Calculate $f'(x)$, then draw the table of variations of f. *(1 pt)*
3. State the value of the maximum of f, then solve $f(x) = 0$. *(1 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $(0, +\infty)$. *(1 pt)*

Exercise 2 · Antiderivatives with a given condition — 5 points

The first four questions are independent. For each, state the expression of the antiderivatives requested and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto 5x^2 - 7x + 3$. *(1 pt)*
2. Determine the antiderivative F of $x \mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0) = 7$. *(1 pt)*
3. Determine all antiderivatives on $\mathbb{R}$ of $x \mapsto \frac{2x}{x^2+7}$. *(1 pt)*
4. Determine an antiderivative on $\mathbb{R}$ of $x \mapsto 2x(x^2+3)^5$. *(1 pt)*
5. Let F and G be two antiderivatives of the same function on an interval I. Prove that there exists a real constant C such that $F(x) - G(x) = C$ for all $x \in I$. *(1 pt)*

Test 4 · Practice 2 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0, +\infty)$. In questions 2 to 5, denote by y the solution satisfying $y(0) = 16$.

1. Determine the constant solution of (E). *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as t tends to $+\infty$. *(1 pt)*
4. Study the variations of y, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 11$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by Y_k the approximation of $y(kh)$. Give the value of Y_0 and express Y_{k+1} in terms of Y_k and h. *(1 pt)*

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, consider $A(7,0,0)$, $B(0,7,0)$ and $C(0,0,7)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane (ABC). *(1 pt)*
2. Let H be the orthogonal projection of O onto (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that OH is the distance from O to the plane (ABC): for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle ABC. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

Test 5 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, we set $I_n = \displaystyle\int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. *(1 pt)*
2. Using integration by parts, calculate I_1. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence (I_n) is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). *(1 pt)*
5. Using integration by parts, express I_{n+1} in terms of I_n and n. *(1 pt)*

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0, +\infty)$. In questions 2 to 5, denote by y the solution satisfying $y(0) = 12$.

1. Determine the constant solution of (E). *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as t tends to $+\infty$. *(1 pt)*
4. Study the variations of y, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 9$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by Y_k the approximation of $y(kh)$. Give the value of Y_0 and express Y_{k+1} in terms of Y_k and h. *(1 pt)*

Test 5 · Practice 1 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, consider $A(9,0,0)$, $B(0,9,0)$ and $C(0,0,9)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane (ABC). *(1 pt)*
2. Let H be the orthogonal projection of O onto (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that OH is the distance from O to the plane (ABC): for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle ABC. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

We set $n = 17$. Consider n independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{5}$. We set $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the distribution followed by S and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 17 variables by 17 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

Test 5 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete attempts will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, we set $I_n = \displaystyle\int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{3x}$.

1. Calculate I_0. *(1 pt)*
2. Using integration by parts, calculate I_1. *(1 pt)*
3. Prove, by comparing the integrands, that the sequence (I_n) is positive and decreasing. *(1 pt)*
4. Show that $I_n \leq \frac{e^3}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). *(1 pt)*
5. Using integration by parts, express I_{n+1} in terms of I_n and n. *(1 pt)*

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{2}y + 5$ on $[0, +\infty)$. In questions 2 to 5, denote by y the solution satisfying $y(0) = 15$.

1. Determine the constant solution of (E). *(1 pt)*
2. Determine the expression of the solution satisfying the given initial condition. *(1 pt)*
3. Determine the limit of $y(t)$ as t tends to $+\infty$. *(1 pt)*
4. Study the variations of y, then determine the smallest real number $t \geq 0$ such that $y(t) \leq 11$. *(1 pt)*
5. Apply Euler's method with step size $h > 0$ and denote by Y_k the approximation of $y(kh)$. Give the value of Y_0 and express Y_{k+1} in terms of Y_k and h. *(1 pt)*

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, consider $A(13,0,0)$, $B(0,13,0)$ and $C(0,0,13)$. We are given $\vec{n} = (1,1,1)$. Lengths, areas and volumes will be expressed in units of length, area and volume respectively.

1. Prove that $\vec{n}$ is normal to the plane (ABC). *(1 pt)*
2. Let H be the orthogonal projection of O onto (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. *(1 pt)*
3. Prove that OH is the distance from O to the plane (ABC): for every $P \in (ABC)$, prove that $OP^2 \geq OH^2$. *(1 pt)*
4. Calculate the area of triangle ABC. *(1 pt)*
5. Calculate the volume of the tetrahedron $OABC$ in two ways, by choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

We set $n = 14$. Consider n independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{4}$. We set $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the distribution followed by S and calculate $E(S)$. *(1 pt)*
2. Using the variance of a Bernoulli random variable and independence, calculate $V(S)$. *(1 pt)*
3. Calculate $E(G)$ and $V(G)$. *(1 pt)*
4. We set $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. *(1 pt)*
5. For this question only, we replace the 14 variables by 14 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. *(1 pt)*

Mock 2 · Practice 1

PRACTICE PAPER · TRAINING

Duration: 4 hours · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, let $I_n = \displaystyle\int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{5x}$.

1. Calculate I_0. *(0.5 pt)*
2. Using integration by parts, calculate I_1. *(0.5 pt)*
3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^5}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). *(1 pt)*
5. Using integration by parts, express I_{n+1} in terms of I_n and n. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into N intervals of equal length, with $N \geq 1$. Write the sums L_N and U_N of the areas of the rectangles built using the values of e^{5x} at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. *(1 pt)*

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{4+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of f. *(0.5 pt)*
3. State the value of the maximum of f, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of f on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of f on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · Practice 1 · continued

Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 10$ and the point $M(3, 0, 0)$. The line d passes through M and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of d. *(0.5 pt)*
2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. *(0.5 pt)*
3. Justify that H is the orthogonal projection of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. *(0.5 pt)*
7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 20 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

Mock 2 · Practice 2

PRACTICE PAPER · TRAINING

Duration: 4 hours · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete approaches will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, let $I_n = \displaystyle\int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. *(0.5 pt)*
2. Using integration by parts, calculate I_1. *(0.5 pt)*
3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. *(0.5 pt)*
4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). *(1 pt)*
5. Using integration by parts, express I_{n+1} in terms of I_n and n. *(0.5 pt)*
6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. *(0.5 pt)*
7. Split $[0,1]$ into N intervals of equal length, with $N \geq 1$. Write the sums L_N and U_N of the areas of the rectangles built using the values of e^{4x} at the left and right endpoints. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. *(0.5 pt)*
8. Find the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. *(1 pt)*

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{3+\ln x}{x}$. On note $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of f. *(0.5 pt)*
3. State the value of the maximum of f, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Determine an antiderivative of f on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of f on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-1) + \ln(x+1) = \ln(63)$, after stating the set on which this equation is defined. *(1 pt)*

Mock 2 · Practice 2 · continued

Exercise 3 · Line, plane and orthogonal projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 6$ and the point $M(1, 0, 0)$. The line d passes through M and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of d. *(0.5 pt)*
2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. *(0.5 pt)*
3. Justify that H is the orthogonal projection of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. *(0.5 pt)*
7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

Exercise 4 · Repeated trials and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. We now repeat the same trial $N \geq 1$ times, again independently. Find the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. A series of 17 trials is accepted if its number of successes is between 1 and 3, inclusive. Calculate the probability that a series is accepted. *(0.5 pt)*
8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0.95$. Run the function and check the minimality of the result. *(1 pt)*

Final mock · Practice 1

PRACTICE PAPER · TRAINING

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of f at 0^+ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of f. *(0.5 pt)*
3. State the maximum value of f, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^1$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of f on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of f on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-2) + \ln(x+2) = \ln(32)$, after stating the set on which this equation is defined. *(1 pt)*

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 9 - 3\cos\left(\frac{\pi t}{5}\right)$, where t is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 10]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of h on $[0, 10]$. *(0.5 pt)*
3. Find all times $t \in [0, 10]$ such that $h(t) = \frac{21}{2}$. *(0.5 pt)*
4. Calculate the mean value of h on $[0, 10]$. *(1 pt)*
5. Study the convexity of h on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(10) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · Practice 1 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 11$ and the point $M(2, 0, 0)$. The line d passes through M and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of d. *(0.5 pt)*
2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. *(0.5 pt)*
3. Justify that H is the orthogonal projection of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. *(0.5 pt)*
7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

Exercise 4 · Repetitions and estimation — 5 points

We repeat 20 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{1}{5}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For N independent trials with the same parameter p, let F_N denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of N and p. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size N so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*

Final mock · Practice 2

PRACTICE PAPER · TRAINING

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete working will be taken into account. Give exact values; round probabilities to 10^{-6} and other approximations to 10^{-4}, unless stated otherwise.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $(0, +\infty)$ by $f(x) = \frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ denote its graph.

1. Determine the limits of f at 0^+ and at $+\infty$. *(0.5 pt)*
2. Calculate $f'(x)$, then draw up the table of variations of f. *(0.5 pt)*
3. State the maximum value of f, then solve $f(x) = 0$. *(0.5 pt)*
4. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$. *(1 pt)*
5. Prove that the equation $f(x) = \frac{1}{2}e^4$ has exactly two solutions in $(0, +\infty)$. *(0.5 pt)*
6. Find an antiderivative of f on $(0, +\infty)$. *(0.5 pt)*
7. Calculate the mean value of f on $[1, e]$. *(0.5 pt)*
8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(40)$, after stating the set on which this equation is defined. *(1 pt)*

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t) = 12 - 4\cos\left(\frac{\pi t}{4}\right)$, where t is expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t \in [0, 8]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights over one revolution. *(0.5 pt)*
2. Calculate $h'(t)$ and draw up the table of variations of h on $[0, 8]$. *(0.5 pt)*
3. Find all times $t \in [0, 8]$ such that $h(t) = 14$. *(0.5 pt)*
4. Calculate the mean value of h on $[0, 8]$. *(1 pt)*
5. Study the convexity of h on the interval of study and determine the coordinates of the points of inflection of its graph. *(0.5 pt)*
6. Determine the total time during which the height is at least equal to the level in question 3. *(0.5 pt)*
7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, and the times at which they are reached. *(0.5 pt)*
8. Calculate the vertical displacement $h(8) - h(0)$, then the sum of the height travelled on the way up and that travelled on the way down. Explain why these two results are different. *(1 pt)*

Final mock · Practice 2 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 7$ and the point $M(3, 0, 0)$. The line d passes through M and has direction vector $\vec{n} = (1, 2, 2)$.

1. Give a parametric representation of d. *(0.5 pt)*
2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. *(0.5 pt)*
3. Justify that H is the orthogonal projection of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. *(0.5 pt)*
4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. *(1 pt)*
5. Check that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ passing through M. *(0.5 pt)*
7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. *(0.5 pt)*
8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. *(1 pt)*

Exercise 4 · Repetitions and estimation — 5 points

We repeat 17 times a trial with two outcomes, called success and failure. The trials are independent and the probability of success is $p = \frac{3}{10}$ on each trial. Let X denote the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. *(0.5 pt)*
2. Calculate the probability of obtaining exactly two successes. *(0.5 pt)*
3. Using complementary events, express and then calculate the probability of obtaining at least two successes. *(0.5 pt)*
4. Calculate the expected value and the variance of X. *(1 pt)*
5. We now repeat the same trial a number $N \geq 1$ of times, still independently. Determine the smallest integer N for which the probability of obtaining at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. *(0.5 pt)*
6. For an integer $j \in \{0, \dots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of the successes. *(0.5 pt)*
7. For N independent trials with the same parameter p, let F_N denote the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ in terms of N and p. *(0.5 pt)*
8. Using the Bienaymé–Chebyshev inequality, determine a sufficient size N so that $P(|F_N - p| < 0.02) \geq 0.95$. Is the size obtained necessarily the smallest possible for this probability? Justify. *(1 pt)*